DIY Serpentine Optical Interferometer
Polarization-Induced Pattern Formation

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GitHub: github.com/Chur-chin/serpentine-optical-interferometer
GitHub: github.com/Chur-chin/Anisotropic-Polarization-Filtering-

Abstract

We report a series of DIY experiments using serpentine-structured waveguides combined with polarization films to observe optical interference and pattern formation phenomena. Three distinct configurations were investigated: (1) a perforated serpentine with z-axis displacement of an output polarization film, (2) a honeycomb serpentine with combined input privacy film and output polarization film, and (3) a rhombus-to-square output geometry with variable air-gap between two polarization films. Each configuration produced qualitatively distinct optical patterns, ranging from blinking and Moiré-to-fringe transitions to global polarization cluster formation.

1. Introduction

Serpentine microstructures have attracted growing interest as platforms for light manipulation, waveguiding, and interference-based sensing. Low-cost, DIY implementations using commercially available polarization and privacy films offer an accessible route to exploring complex optical phenomena. In this work, we systematically characterize pattern formation as a function of structural geometry and film configuration.

2. Experimental Configurations

2.1 Perforated Serpentine — Z-axis displacement of output polarization film

Setup: A serpentine structure with perforations (open channels) was used. A single polarization film was placed at the output side and translated along the z-axis.

Observations:

• At small z-displacements: **blinking** behavior was observed, characterized by periodic intensity modulation.

• As displacement increased: a clear **Moiré** $\rightarrow$ **Fringe transition** occurred. The broad, low-spatial-frequency Moiré pattern gave way to high-contrast linear fringe patterns, consistent with far-field interference of diffracted beams from the periodic serpentine apertures.

Interpretation: The perforations act as a periodic aperture array. Moving the polarization film in z changes the effective path length and angular filtering condition, driving the system from near-field Moiré overlap into a far-field fringe regime.

2.2 Honeycomb Serpentine — Privacy film (input) + Polarization film (output)

Setup: A honeycomb-geometry serpentine was used. A privacy protection film (angular light-restricting) was placed at the input side; a polarization film at the output side.

Observations: Clear **discrete pattern** formation was confirmed. The honeycomb lattice symmetry was imprinted onto the transmitted intensity distribution, producing a spatially quantized cluster arrangement rather than a continuous fringe pattern.

Interpretation: The privacy film restricts the angular cone of input illumination, selecting specific Bloch-like modes of the honeycomb lattice. The output polarization film converts polarization-encoded spatial information into intensity contrast, revealing the discrete mode structure.

2.3 Rhombus-to-Square Output Geometry — Air-gap between two polarization films

Setup: The output aperture geometry was switched between rhombus and square. Two polarization films were placed at the output with variable z-separation.

Observations:

Film Separation	Observed Pattern
0 mm (contact)	Local cluster — spatially confined, isolated polarization domains
3 mm air gap	Global cluster — spatially extended, system-wide correlated domains

Interpretation: At contact, birefringent coupling is localized: each domain independently selects a polarization state, producing isolated (local) clusters. A 3 mm gap allows the diffractive field to propagate and mix spatially, producing long-range correlations and a globally coherent cluster pattern.

3. Summary of Results

Configuration	Input	Output	Key Phenomenon
Perforated Serpentine	—	Polarization film (z-scan)	Blinking; Moiré → Fringe
Honeycomb Serpentine	Privacy film	Polarization film	Discrete pattern
Rhombus↔Square	—	2× Pol. film (contact)	Local cluster
Rhombus↔Square	—	2× Pol. film (3 mm gap)	Global cluster

4. Discussion

The three experimental configurations demonstrate a progression from single-film, single-variable experiments (z-axis blinking/fringe) to dual-film, geometry-dependent cluster formation. The key finding is that **spatial separation between polarization films acts as a tuning parameter for cluster correlation length**: contact → local, 3 mm → global. This is analogous to a coupling-length-dependent phase transition in condensed matter systems, suggesting that serpentine optical platforms may serve as table-top analogues for studying symmetry breaking and cluster formation.

The honeycomb result further indicates that input-side angular filtering (privacy film) is sufficient to impose lattice symmetry on the output pattern — a result with potential implications for structured-light generation and optical mode selection.

5. Conclusion

We have demonstrated three DIY serpentine optical interferometer configurations:

1. **Perforated serpentine + z-scan output film** → Blinking and Moiré-to-Fringe transition

2. **Honeycomb serpentine + privacy/polarization films** → Discrete pattern formation

3. **Rhombus-square output + dual polarization films** → Local cluster (contact) and Global cluster (3 mm gap)

These results establish a simple, accessible experimental framework for exploring polarization-structured light and cluster formation in periodic microgeometries. Full experimental data and code are available at the GitHub repositories listed above.

References

[1] DIY Serpentine Optical Interferometer: https://github.com/Chur-chin/serpentine-optical-interferometer

[2] Anisotropic Polarization Filtering: https://github.com/Chur-chin/Anisotropic-Polarization-Filtering-